Muhammad Maikudi ISAH

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EDUCATION

Jan 2019–May 2022 Doctor of Philosophy, Physics, University of Parma, Italy.
Thesis title: "Muon sites and couplings in magnetic and superconducting materials: towards high-throughput modelling".
Supervisor: Prof. Roberto De Renzi and Dr. Ifeanyi John Onuarah

Sep 2017–Aug 2018 Postgraduate Diploma Programme, Condensed Matter Physics
The Abdus Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy.
Thesis title: "Understanding force-charge correlations in bulk water".
Supervisor: Prof. Ali Hassanali, Prof. Alessandro Laio, and Dr. Emiliano Poli
Note: This is a one-year pre-PhD programme with 12 courses and a 3-month research project.

Jul 2016–Dec 2017 Master of Science, Theoretical and Applied Physics
African University of Science and Technology (AUST), Abuja, Nigeria.
Thesis title: "Tight-binding descriptions of graphene and its derivatives".
Supervisor: Dr. Emine Kucukbenli
Note: This is an 18-month intensive Masters programme consisting of 14 courses and an independent research project.

Jun 2012–Feb 2016 Bachelor of Science, Physics, Bayero University Kano (BUK), Kano, Nigeria.
Thesis title: "Numerical simulation of the radioactive decay law of Uranium-238 using MATLAB program".
Supervisor: Prof. Abdulkadir Muhammad Nura

EXPERIENCE

Nov 2022–Oct 2025 Post-doctoral Fellowship
Department of Physics and Astronomy, University of Bologna, Italy.
Project title: "Probing novel quantum materials with strong spin-orbit coupling by using nuclei and muons as local probes".
Supervisor: Dr. Samuele Sanna

MAJOR RESEARCH PROJECTS AND COLLABORATIONS

- FAIR automated workflows (AiiDa-Muon) for high-throughput computational study of the muon embedding for the investigation of solid state materials. Here, I contributed to the early-stage development of the AiiDA-Muon code, particularly in its methodology and software implementation.
- Software development for atomistic and muon simulation (in collaboration with Dr. Pietro Bonfà of the μSR and NMR group, UNIPR). Here, I have co-developed and tested the UNDI code.
- Combined DFT (+DMFT/Hubbard-I model) calculations and experimental measurements to investigate the magnetic behavior and the role of spin-orbit-coupled interactions in Re-based double perovskites.
- Study of muon effects in superconductors with time reversal symmetry breaking, where spontaneous magnetization is observed with μSR (in collaboration with the group of Prof. Tom Lancaster of the Durham University, UK and Prof. Stephen Blundell of the Oxford University, UK).
- Role of exchange interations in the spin dynamics of Tb-based molecular nanomagnets (in collaboration with the group of Prof. Alessandro Lascialfari of the Universitá di Pavia, Italy, the group of Prof. Loranzo Sorace of the Universitá di Firenze, Italy and Prof. Zaher Salman of the PSI Center for Neutron and Muon Sciences, Switzerland).

PUBLICATIONS

[1] **Isah M. M.**, Dalal B., Kang X., Fiore Mosca D., Onuorah I. J., Scagnoli V., Bonfà P., De Renzi R., Belik A. A., Franchini C., Yamaura K., Sanna S., "Magnetic behavior of the $5d^1$ Re-based double perovskite $\mathrm{Sr_2ZnReO_6}$", Phys. Rev. B **113**, 104431 (2026).

[2] Onuorah I. J., Bonacci M., **Isah M. M.**, Mazzani M., De Renzi R., Pizzi G., Bonfà P., "Automated computational workflows for muon spin spectroscopy", Digital Discovery **4**, 523 (2025).

[3] Casadei M., **Isah M. M.**, Cabassi R., Trevisi G., Fabbrici S., Belli M., de Julián Fernández C., Allodi G., Fournée V., Sanna S., Albertini F., "Magnetic properties of Ge, Re and Cr substituted $\mathrm{Fe_5SiB_2}$", Journal of Alloys and Compounds **1026**, 180346 (2025).

[4] **Isah M. M.**, Sorace L., Lascialfari A., Arosio P., Salman Z., Nogueira A. M., Poneti G., Mariani M., Sanna S., "μSR evidence of a marked exchange-interaction effect on the local spin dynamics of Tb-based molecular nanomagnets", Phys. Rev. B **111**, 014444 (2025).

[5] Onuorah I. J., Frassineti J., Wang Q., **Isah M. M.**, Bonfà P., Rau J. G., Rodriguez-Rivera J. A., Kolesnikov A. I., Mitrovi ć V. F., Sanna S., Plumb K. W., "Unraveling the magnetic ground state in the alkali-metal lanthanide oxide $\mathrm{Na_2PrO_3}$", Phys. Rev. B **110**, 064425 (2024).

[6] **Isah M. M.**, Renzi R. D., Onuorah I. J., "Magnetic properties and muon localization in $\mathrm{Cr_2S_3}$", Journal of Physics: Conference Series **2462**, 012006 (2023).

[7] Bonfà P., Frassineti J., Wilkinson J. M., Prando G., **Isah M. M.**, Wang C., Spina T., Joseph B., Mitrovi ć V. F., De Renzi R., Blundell S. J., Sanna S., "Entanglement between Muon and $I > \frac{1}{2}$ Nuclear Spins as a Probe of Charge Environment", Phys. Rev. Lett. **129**, 097205 (2022).

[8] Onuorah I. J., **Isah M. M.**, De Renzi R., Bonfà P., "Frustrated network of indirect exchange paths between tetrahedrally coordinated Co in $\mathrm{Ba_2CoO_4}$", Phys. Rev. Mater. **5**, 124407 (2021).

[9] Huddart B. M., Onuorah I. J., **Isah M. M.**, Bonfà P., Blundell S. J., Clark S. J., De Renzi R., Lancaster T., "Intrinsic Nature of Spontaneous Magnetic Fields in Superconductors with Time-Reversal Symmetry Breaking", Phys. Rev. Lett. **127**, 237002 (2021).

[10] Bonfà P., **Isah M. M.**, Frandsen B. A., Gibson E. J., Brück E., Onuorah I. J., De Renzi R., Allodi G., "Ab initio modeling and experimental investigation of $\mathrm{Fe_2P}$ by DFT and spin spectroscopies", Phys. Rev. Mater. **5**, 044411 (2021).

[11] Bonfà P., Frassineti J., **Isah M. M.**, Onuorah I. J., Sanna S., "UNDI: An open-source library to simulate muon-nuclear interactions in solids", Computer Physics Communications **260**, 107719 (2021).

AWARDS, DISTINCTIONS AND FELLOWSHIPS

- 2020: ISMS Young Scientist Best Poster Award, 15th International Conference on Muon Spin Rotation (μSR2020), University of Parma, Italy. Awarded for development of a high-throughput DFT framework for muon-site calculations.
- 2019: Italian Ministry of Education PhD Scholarship (*Borsa di Studio*).
- 2017: UNESCO/IAEA Study Grant, Italy.
- 2017: Best Master's Graduate Award, AUST, Abuja, Nigeria.
- 2016: Full Scholarship for Master's Degree Studies.
- 2016: Prof. G.G. Parfitt Prize for Best Final Year Physics Student.
- 2013–2016: Best Physics Department Student Awards.
- 2011: Third Place, Nigerian Mathematics and Science Olympiad in Chemistry, State Competition, Kano, Nigeria.

SELECTED CONFERENCES, TALKS AND PRESENTATIONS

- Jul 2025: "*μSR evidence of a marked exchange-interaction effect on the local spin dynamics of Tb-based molecular nanomagnets*", FisMat2025, Venezia, Italy – Oral Presentation
- Jun 2025: International Conference on Magnetism (ICM2024), Bologna, Italy – Oral Presentation
- Feb 2024: "*Anomalous Cascade of Transitions in Sr_2ZnReO_6 with Strong Spin-Orbit Coupling*", MAGNET2024–VIII Italian Conference on Magnetism, Milan, Italy – Poster Presentation
- Feb 2024: "*Anomalous Spin Dynamics of Tb-based Molecular Nanomagnets*", MAGNET2024–VIII Italian Conference on Magnetism, Milan, Italy – Poster Presentation
- Aug 2022: "*Calculating muon sites and couplings from a high-throughput modelling perspective*", 15th International Conference on Muon Spin Rotation, Parma, Italy – Poster and Oral Presentation
- Feb 2021: "*High-throughput muon site modelling*", 7th Italian Conference on Magnetism, Florence, Italy, (remote) – Oral Presentation
- Sep 2020: "*High-throughput characterization of muon sites*", μSR Site Calculation Meeting 2020, United Kingdom, (remote) – Oral Presentation
- Feb 2020: "*Predicting muon embedding sites and fields in selected magnetic material*", 5th Italian School on Magnetism, Rome, Italy – Poster Presentation
- Jul 2019: International Advanced School in Muon Spectroscopy, ISIS, Didcot, United Kingdom
- Sep 2019: Advanced School on Neutrons and Muons for Magnetism, Ispra (Varese), Italy
- Sep 2019: Autumn School on Correlated Electrons: Many-Body Methods for Real Materials, Forschungszentrum Jülich, Germany
- Feb 2019: 15th Advanced School on Parallel Computing, Bologna, Italy

COMPUTATIONAL GRANTS

- STFC SCARF Cluster 2019/2025 (Collaborative access via Dr. Stephen Cottrell, Rutherford Appleton Laboratory).
- SUPER 2020/2021 CINECA (660,000 CPU hours†), "High-throughput characterization of muon sites".

- ISCRA 2019/2020 Type C (55,000 CPU hours†), "DFT Search of Muon Localization Sites and Induced Perturbations in Selected Magnetic and Superconducting Systems".

† The grant was awarded as principal investigator.

SKILLS

Tools	Python, Fortran, MATLAB, Linux, Git, LaTeX, VESTA, Origin, Gnuplot, XCrySDen
Frameworks	Quantum ESPRESSO, CP2K, AiiDA, aiida-muon, musrfit, mulab, WiMDA (Basic), UNDI
Languages	English (Proficient), Italian (Intermediate), Hausa (Native)

REFERENCES

1. Prof. Samuele Sanna
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2. Prof. Roberto De Renzi
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3. Dr. Ifeanyi John Onuarah
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4. Dr. Pietro Bonfà
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